

Nursery Maintenance Guidebook

**Background and instructions for how to clean an
in-water coral nursery.**



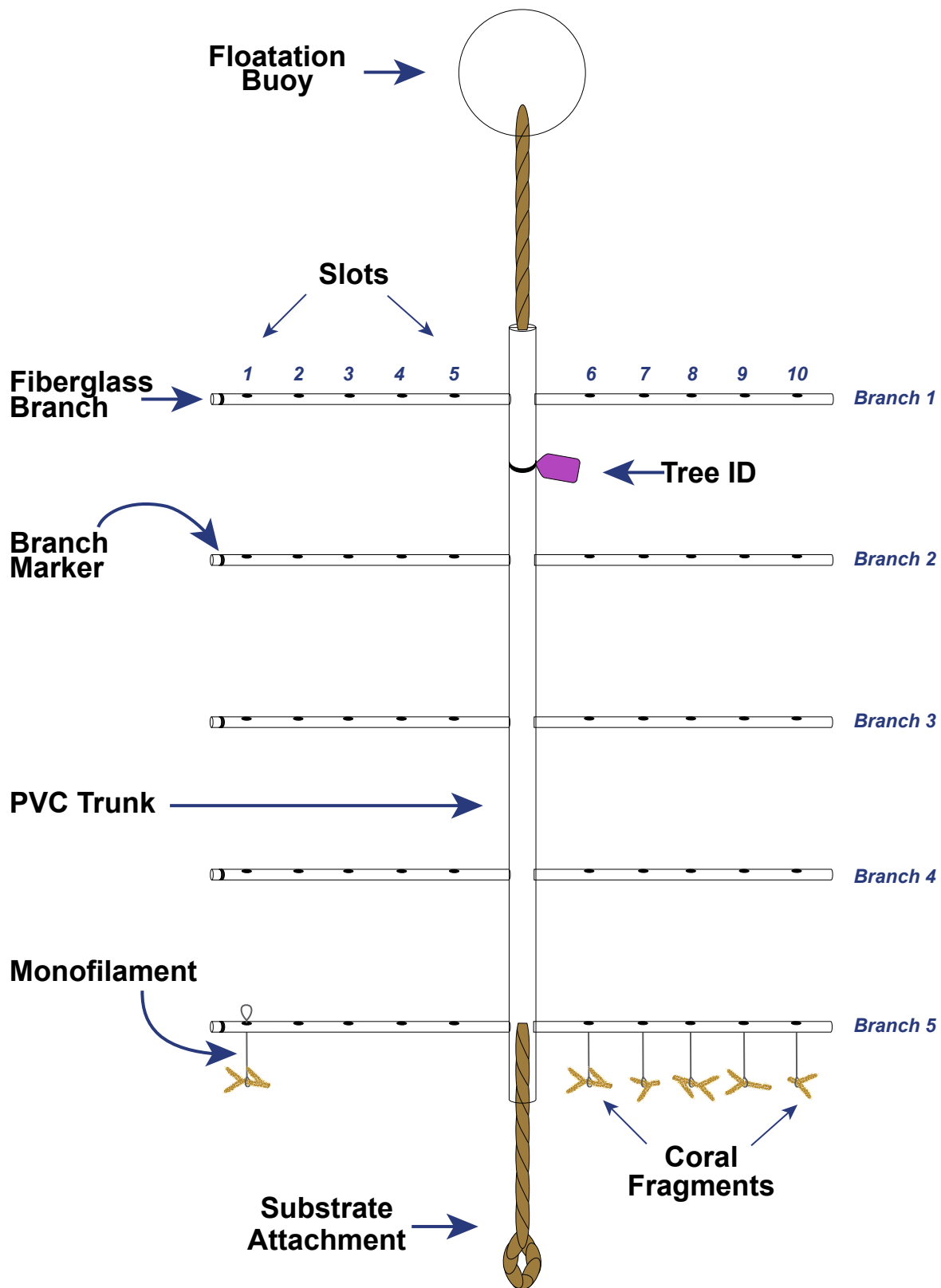
**Presented By:
Reef Response**

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Tree Diagram



Maintenance

Supplies list:

- Scrub pads (2)
- Wire brush (1)
- Dive knife/scrapper
- Gloves
- Optional: Camera (1 per buddy team)
- Optional: Clipboard and datasheet (1 per buddy team)
- For Boat: Vinegar (to alleviate fire coral stinging)



Cleaning Instructions

Trees should be cleaned in dive buddy teams. Before getting to work on scrubbing the trees, go through in-water buddy check to make sure all equipment is functioning properly and streamlined, establish buoyancy control, and familiarize yourself with the site.

CAUTION DISCLAIMER

The most important task when diving is to be safe, everything else is secondary! It is easy to become tangled in the corals and branches. Please be sure to use caution for your safety and the well-being of the corals.

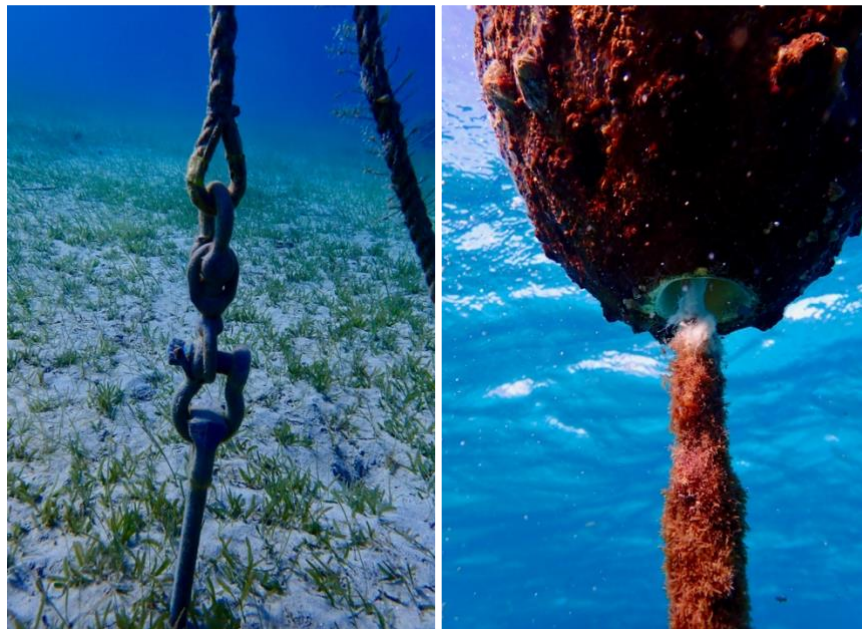
Prior to Cleaning

1. In-water Buddy Check

- a. Hair – long hair needs to be secured and not flowing to avoid entanglement.
- b. Equipment attached to BCD.
- c. Gear streamlined and not loose hanging (e.g., octopus regulator and pressure gauge).
- d. Buoyancy – establish neutral buoyancy to maintain consistent depth.

2. Assess the overall condition of the coral nursery tree.

- a. Before starting the cleaning process, evaluate the condition of the tree. Look for any signs of damage, disease (see pg. 10), or excessive algae growth.
- b. Inspect the hardware at the sand attachment and the float for abrasion or damage that may cause a tree to detach. If noticeable hardware damage is present, please take a picture and inform the divemaster so the Reef Response team may be notified.



3. Take Pictures

- a. Take picture of the **tree number tag** and name tag, located near the top of the tree (e.g. 4274, PROF. OAK).



- b. Take 'before' pictures of the trees to document the current state.

4. (optional) Record

- a. On the data sheet, record your names, **Location**, **Date**, and tree **Tag Number** and name.

Cleaning

The instructions below breakdown the process of cleaning the tree. The components of the tree are listed in order of importance and should be cleaned following the same order. If cleaning a tree in a buddy team, be sure to stay on opposite sides of the tree. Do not clean below a diver as the remnants will sprinkle down on top of you and can become quite itchy and irritating. Clean branch by branch starting from the bottom branch and working your way up.



1. Monofilament

- a. The monofilament attaching the coral fragment to the branch is the most important part to be cleaned as it has direct contact with the coral.
- b. A successful method is to **wrap the scrub pad around the monofilament**, pinch with your fingers and run the scrub **up and down** until clean.



- c. Be sure to clean the top loop above the branch as well. You can pull some of the monofilament through the top of the branch to clean the entire length of monofilament.



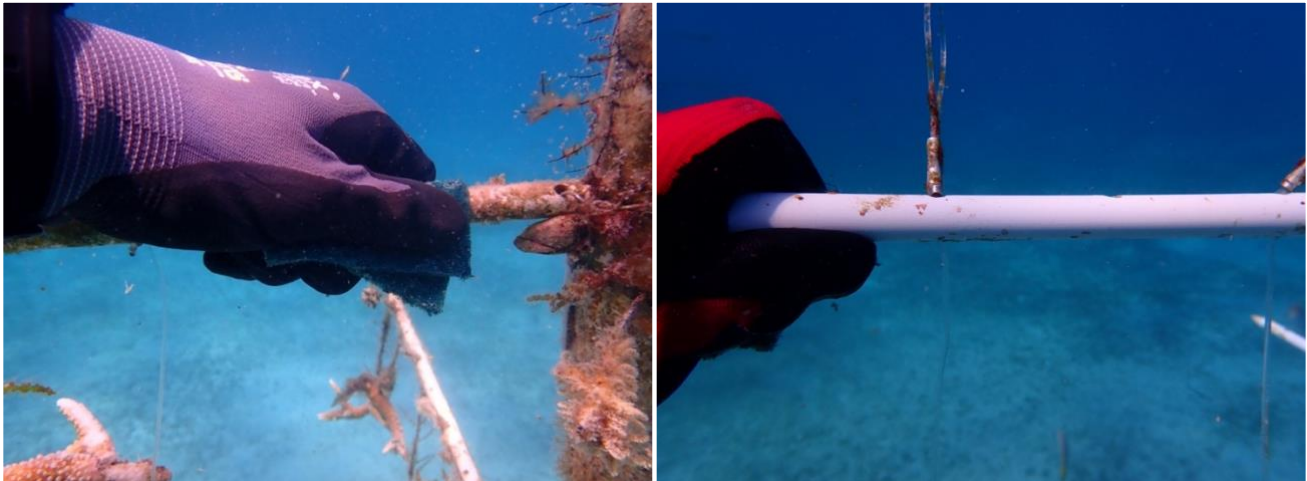
- d. **Do not to pull too hard** on the line so it does not break free. However, they do come unattached occasionally.
 - i. If this does happen place fragment by base of tree and report to Dive lead to record tree number, branch number, and slot number.

2. Algae on Coral Fragment

- a. If there are any algae stuck to the coral, gently remove by hand. **Be careful not to harm the coral.**

3. Fiberglass Branch

- a. Scrubbing the fiberglass will likely be the most intensive aspect. Algae, fire coral, clams, and other organisms may be attached to the branch (see pg. 11). **Be sure to wear your gloves!**
- b. Begin scrubbing with the scrub pad to remove the algae until clear.



- c. For hard to remove **encrusting algae / fire coral**, use the **wire brush** to clean the branch.
 - i. ***WARNING***: Fire coral may cause slight stinging sensations on one's face when being removed. This is generally not a cause for concern unless you have an allergy. **Vinegar** can help alleviate the ailment if severe or painful stinging occurs. If you experience a more severe reaction, immediately communicate to your buddy that you are in distress, return to the boat in a safe manner, and assess the situation with your dive team to determine if further medical action is needed.
- d. Barnacles and clams are often easiest to remove with a dive knife / scraper. They will pop off if you can get underneath an edge. The fiberglass can splinter so **be careful** not to remove the paint layer. Fiberglass splinters can penetrate the gloves and **do not feel good**. When scraping with a knife or scraper, scrape away from yourself!



4. PCV Trunk

- a. Wrap the scrub pad around the PCV and clean with an up and down twisting motion to get all the way around.



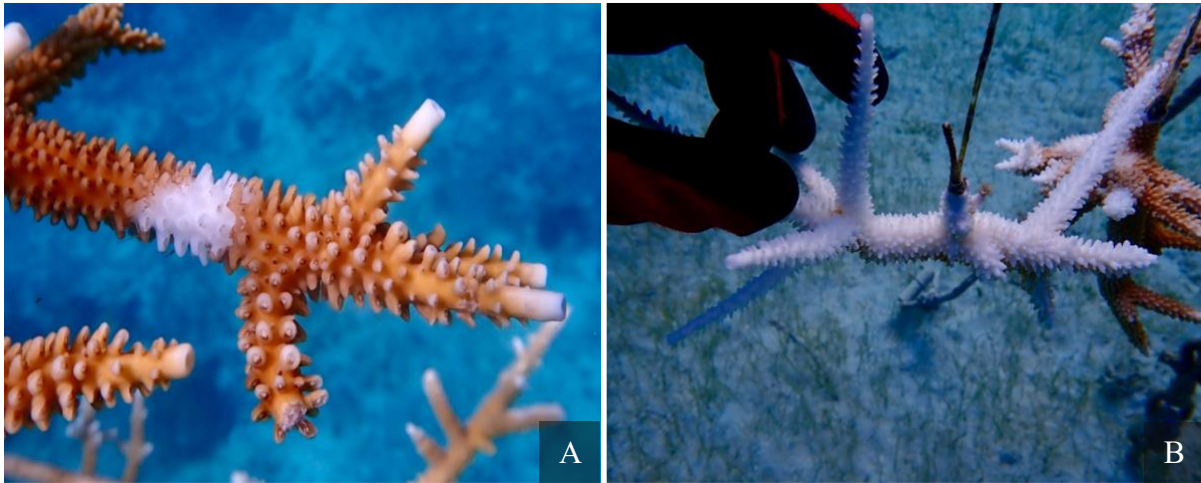
5. Buoy

- a. If time permits, remove barnacles and other fouling from the floatation buoys with a dive knife / scraper.



Coral Health

Part of maintaining our coral nurseries involves doing health assessments of the coral fragments to make sure they are healthy and growing. It can take years of training and expertise to properly identify coral health, but recognizing some signs of problems and alerting Reef Response personnel to any issues will help us greatly!



Tissue Loss

The above left photo (A) shows a staghorn coral displaying signs of a health issue. The white area is where coral tissue has been lost and the white skeleton is exposed. This type of tissue loss could be a predation scar or be some sort of rapid tissue loss condition or disease. Disease outbreaks have the potential to spread rapidly and cause significant losses. It is therefore very important to identify signs of rapid tissue loss in the nursery early so action can be taken quickly.

Paling / Bleaching

The above right photo (B) is a staghorn coral showing signs of paling/bleaching. The white areas on this fragment are places where the tissue is still present but has lost its color (compared with tissue loss in the previous section, where the white is an area where there is no longer tissue). It is not necessary to properly identify the differences, but to recognize and photograph irregularities like this and to **report to the Reef Response team**.

Critters You May Encounter

A few additional critters you may encounter during cleaning to be mindful of:

1. **Fire worms** are coral predators and should be carefully removed but do not touch with bare hands.
2. **Crabs** on the trees are algal grazers and should not be removed.
3. **Urchins** may have settled within the floatation buoy.
4. **Amphipods** live within the algae on the trees and tend to cling to wetsuits after the algae is removed. They are not harmful and can be easily wiped off in the water following the dive.
5. **Barracuda** look intimidating but they are not a concern.
6. **Squid** squad.



Thank You!

Reef Response thanks you for all your help! Please reach out to us with any questions, comments, or concerns. Together we can all work towards a brighter future for our wonderful planet.



Contact Us

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